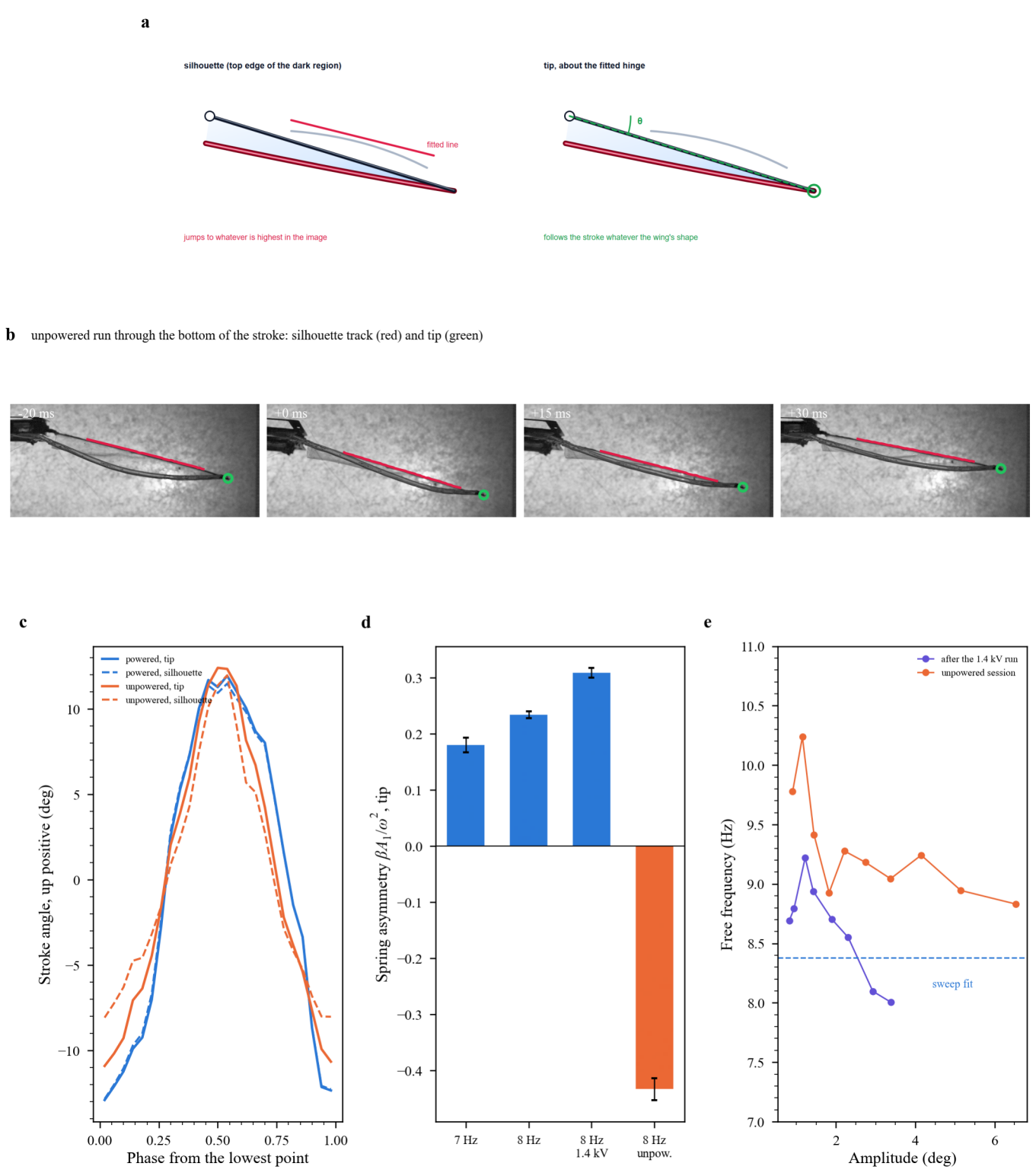

a
silhouette (top edge of the dark region)
fitted line
jumps to whatever is highest in the image
tip, about the fitted hinge
θ
follows the stroke whatever the wing's shape
b unpowered run through the bottom of the stroke: silhouette track (red) and tip (green)
-20 ms
+0 ms
+15 ms
+30 ms
c
powered, tip
powered, silhouette
unpowered, tip
unpowered, silhouette
Stroke angle, up positive (deg)
Phase from the lowest point
d
Spring asymmetry $\beta A_1/\omega^2$, tip
7 Hz
8 Hz
8 Hz 1.4 kV
8 Hz unpow.
e
after the 1.4 kV run
unpowered session
sweep fit
Free frequency (Hz)
Amplitude (deg)